Research Highlights

- FFE-v2 adds re-segmentation to commuting Lie generators of packet-level actions.
- Score-only FFE-v2 matches or exceeds our greedy white-box control in 9 of 9 cells.
- On XGBoost it evades 99.5/100/98.3% of detected attacks; FFE-v1 gets 57.6% on UNSW.
- CatBoost on CIC-IDS2017 stays at 36.4%: immutable features set a ceiling.
- A monotone gate keeps flagged flows flagged only if backward features stay fixed.